CTFS:

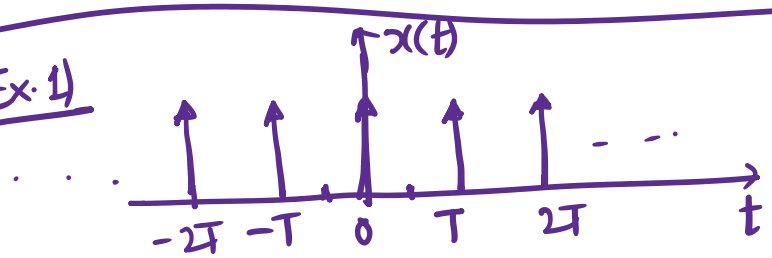
$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow \text{a.e. (not pointwise convergence)}$$

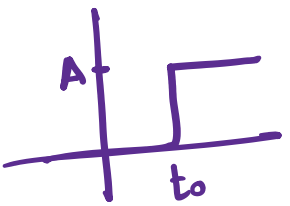
$$a_k = \frac{1}{T}\int_{\langle T\rangle} x(t) e^{-jk\omega_0 t}\, dt$$

Dirichlet Condn: (Sufficient but not Necessary)

1) $\int_{\langle T\rangle} |x(t)|\, dt < \infty$
2) $x(t)$ has finite no. of maxima & minima over a period
3) " " finite discontinuities " "

Ex. 1)



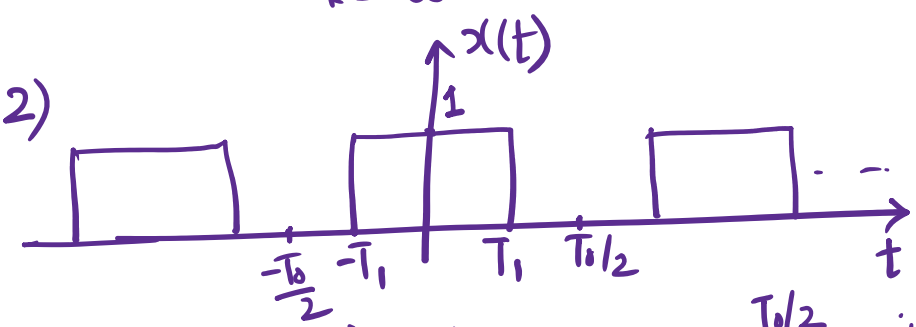


$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}, \quad a_k = \frac{1}{T}\int_{\langle T\rangle} x(t) e^{-jk\omega_0 t}\, dt$$

$$= \frac{1}{T}\int_{\langle T\rangle} \delta(t) e^{-jk\omega_0 t}\, dt = \frac{1}{T} \quad \forall k$$

$$\therefore x(t) = \frac{1}{T}\sum_{k=-\infty}^{\infty} e^{jk\omega_0 t}$$

2)

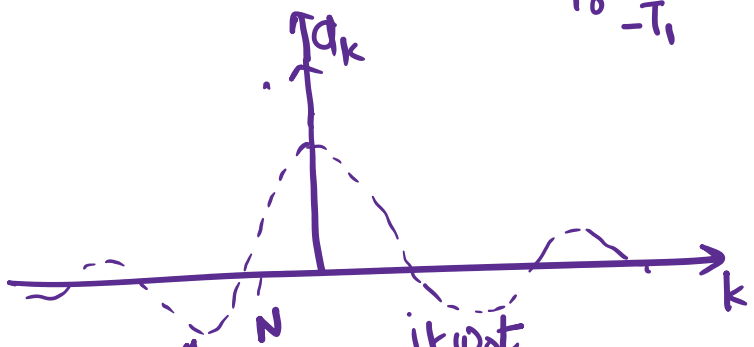


$$x(t) = \sum a_k e^{jk\omega_0 t}, \quad a_k = \frac{1}{T_0}\int^{T_0/2} x(t) e^{-jk\omega_0 t}\, dt$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad , \quad a_k = \frac{1}{T_0}\int_{-T_0/2} x(t) e^{-jk\omega_0 t}\, dt$$

$$= \frac{1}{T_0}\int_{-T_1}^{T_1} 1 \cdot e^{-jk\omega_0 t}\, dt = \frac{e^{-jk\omega_0 t}\Big|_{-T_1}^{T_1}}{-jk\omega_0 \cdot T_0}$$

$$= \frac{\sin k\omega_0 T_1}{k\pi}$$



$$\Rightarrow \quad x_N(t) \triangleq \sum_{k=-N}^{N} a_k e^{jk\omega_0 t}$$

$$\sum_{k=-\infty}^{\infty} |a_k|^2 = \frac{1}{T}\int_{\langle T\rangle} |x(t)|^2\, dt \quad , \quad a_{-k} = a_k^* \text{ (for real } x(t))$$

$$\sum a_k \cdot a_k^* = \sum a_k \cdot \frac{1}{T}\int_{\langle T\rangle} x^*(t) e^{jk\omega_0 t}\, dt$$

$$= \frac{1}{T}\int x^*(t) \underbrace{\sum a_k e^{jk\omega_0 t}}_{x(t)}\, dt = \frac{1}{T}\int_{\langle T\rangle} |x(t)|^2\, dt$$